

Proposed Workflow

Always ask if you have questions

1. Literature Search

- a. An important taxonomic note. Our database is focused on collecting body sizes for genera. In practice, this means that for most genera we only have a set of measurements from a single specimen that belongs to a single species. However, when collecting data we always want to make sure we record which species we are making measurements from. Ideally, we want to measure the holotype specimen of the type species for the genus. However, this is not always practical, so any representative will suffice in those situations. If it's relatively convenient, you can always take measurements from multiple specimens and species in a reference and record them on separate lines.
- b. search genus name in Google Scholar
- c. identify source that seems to have size information
 - sizes can be stated in the text or extracted from figures. to be extracted from a figure, the figure must contain either a magnification (usually stated in caption) or the figure must have a labeled scale bar. If these are lacking the reference is not useful.
 - hint: articles that don't have size info, but do have species authority (name & year after sp. name) are likely to have a reference to the original description.
 - note: watch out for misspellings in our genus names—after searching for a while you may notice that there are some variations on standard name suffices and prefixes. If you think there is a misspelling, search the new spelling and if you think the new spelling is in fact correct, add the new spelling to the “genus_new_spelling” field in the spreadsheet.
- d. copy ref from “Cite” link and paste into spreadsheet “reference” field
- e. if PDF available: download PDF and rename file to the genus name—you may end up with the same article downloaded for multiple genera, but with different names—this is okay.
- f. if nothing could be found, make a note in notes field saying Google Scholar failed.
- g. once you've searched all genera, go back through and repeat search in GeoRef/GeoscienceWorld only for those genera that failed in Google Scholar.

2. Size Measurements and Other Data Extraction

- a. sort spreadsheet on “reference” field

- b. start with genera you have a PDF for
 - i. read through pdf to extract size information from text
 - If sizes are stated in text, convert to **millimeters** and record in spreadsheet. Indicate a figure magnification of 1. **YOU MUST ENTER A MAGNIFICATION.**
 - If sizes are not listed in text, but there is a measurable figure, search for a paper copy of the journal/book at Stanford Library and record call number in “call_number” field. If not in Library, record print in the “call_number” field, and we’ll get it printed for you.
 - extract age and geological information from the text.
 - **Do not enter anything into measurement/magnification fields that is not a number. Make sure all units are in mm.**
- c. measure sizes from hard copies of references (either print outs or library materials)
 - i. measure figures with digital calipers
 - record specimen dimensions from calipers in **millimeters**. make sure calipers are at **0 mm when closed**.
 - if there is a magnification, record that in the spreadsheet. if there is a scale bar, measure it and record the caliper reading in the “scale_bar_measured” field, then record the scale bar label in the “scale_bar_stated” field. Convert the stated scale bar label to millimeters if in other units.
 - **YOU MUST ENTER A MAGNIFICATION or MEASURE A SCALE BAR.**
 - **Do not enter anything into measurement/magnification fields that is not a number. Make sure all units are in mm.**
 - ii. if you are working from a source that you do not have a PDF for, extract age and geological formation data from the text as in part 2b above.
- d. **DO NOT MEASURE ORNAMENTAL SPINES.** We want to measure the central part of the body that contains the metabolically active tissue, so don't include spines, keels, or other shell ornamentation in size measurements. **If in doubt, ask.**

Type specimen codes (leave blank if not measuring a type specimen)

H = holotype

P = paratype

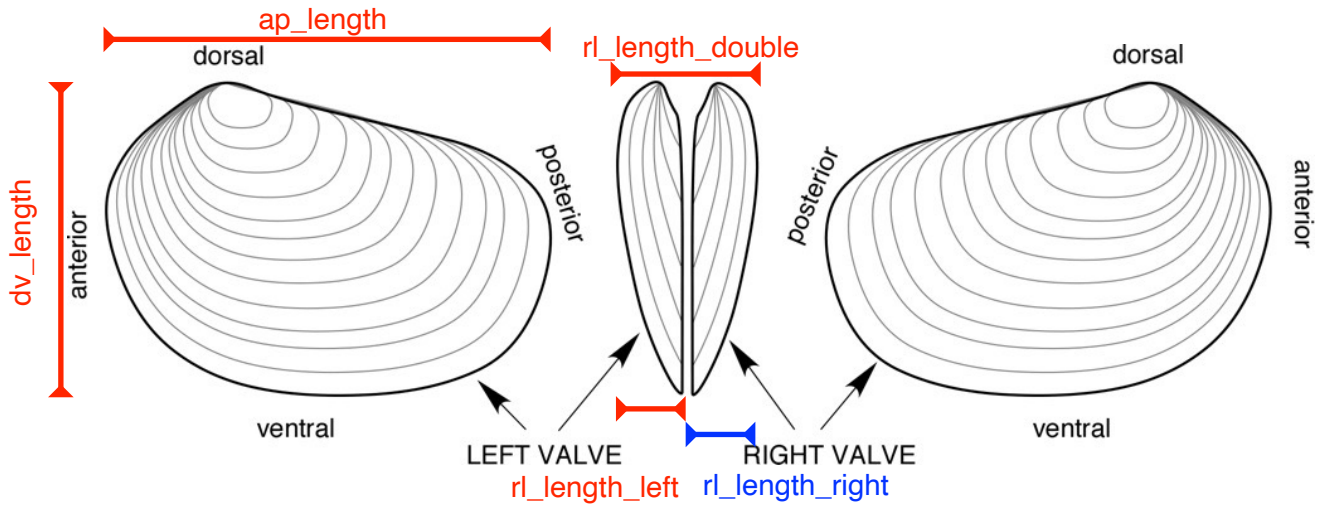
N = neotype

L = lectotype

S = syntype

O = other type of designated type specimen (e.g., paralectotype, hapantotype). Do not use "O" if you are not measuring a type specimen.

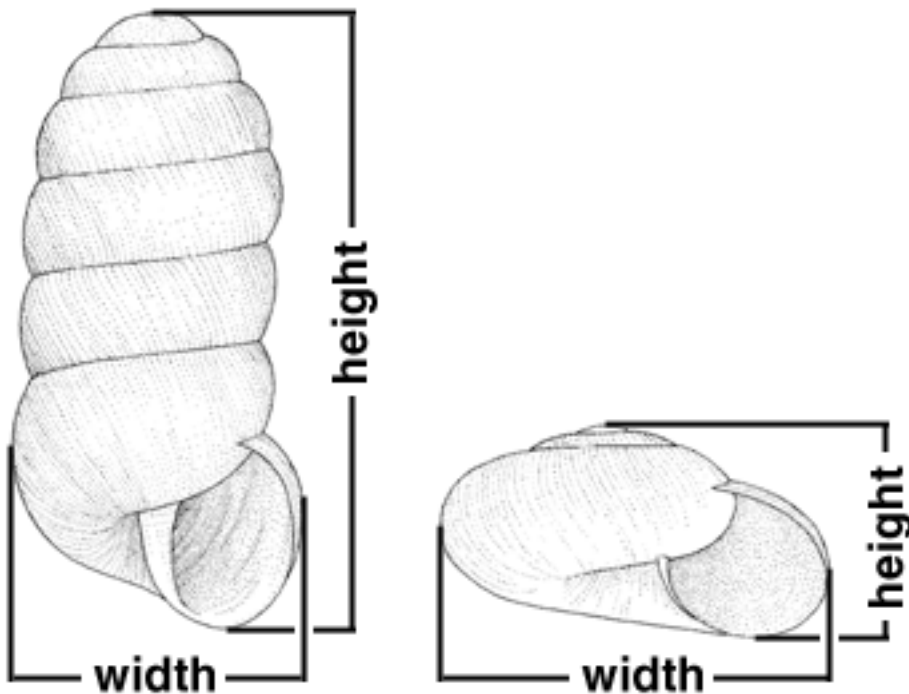
Bivalves



Bivalve Measurements to Make

- ap_length
- dv_length
- rl_length_both (both valves)
- rl_length_right (single right valve, if disarticulated)
- rl_length_left (single left valve, if disarticulated)

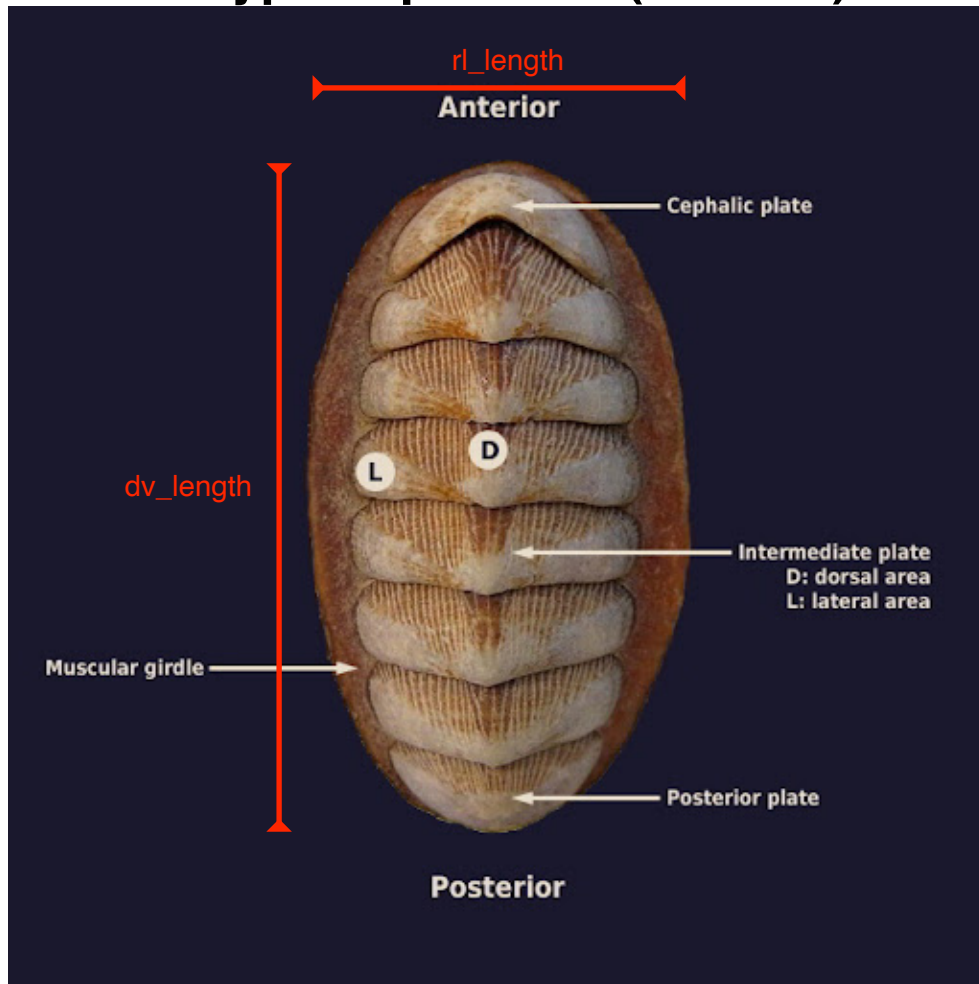
Gastropods



Gastropod Measurements to Make

- height
- width

Polyplacophorans (chitons)



Polyplacophoran Measurements to Make

dv_length

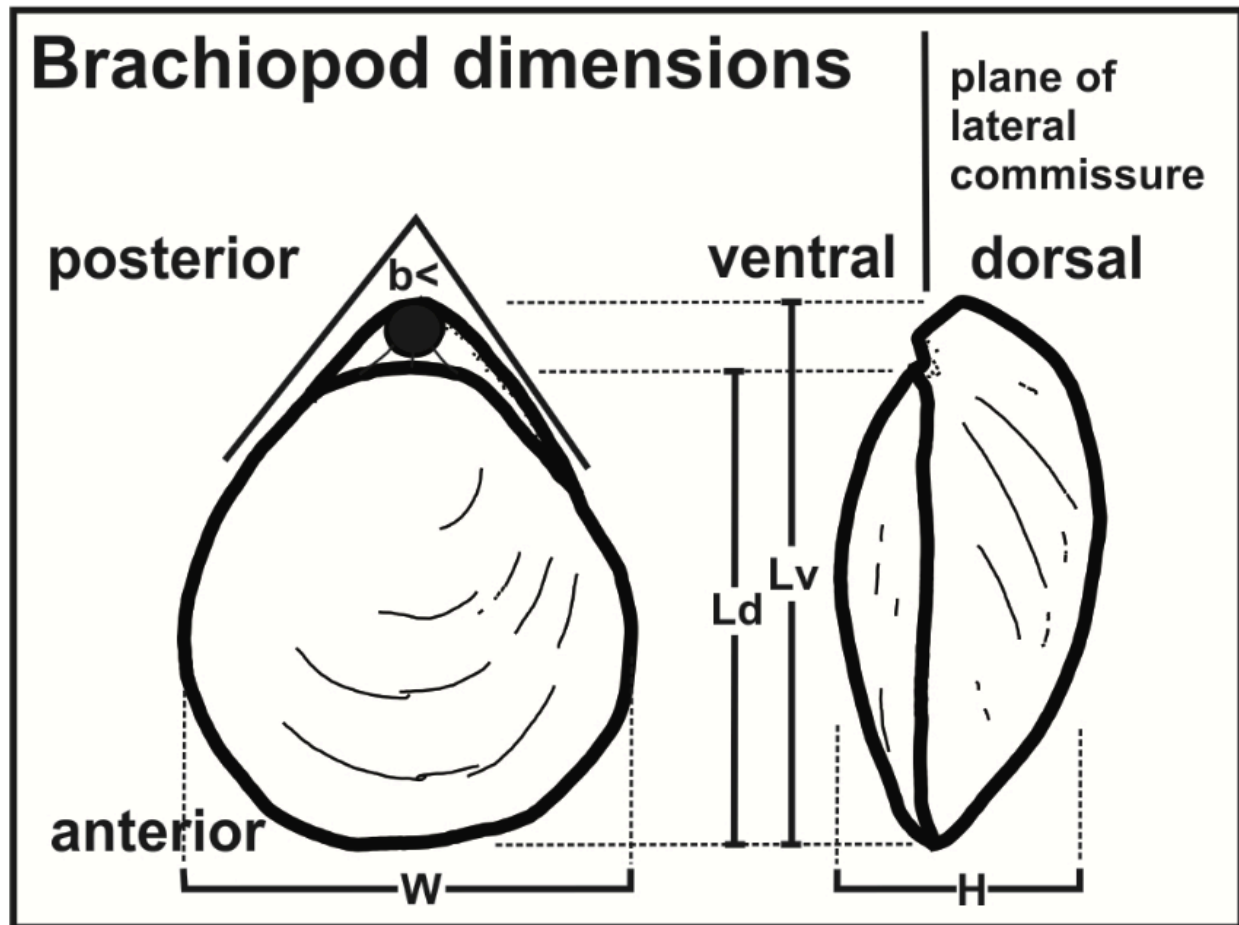
rl_length

dv_length (perpendicular plane of image)

[This needs work to include disarticulated fossil plates]

If you have images that only include the plates and not the muscular girdle, let me know!

Brachiopods



Brachiopod Measurements to Make

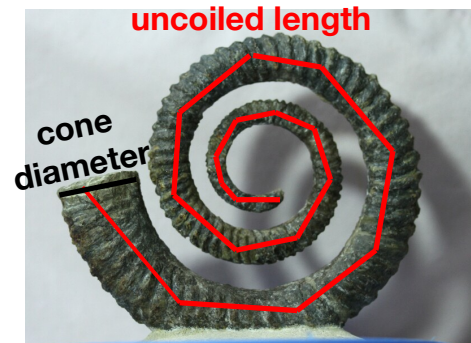
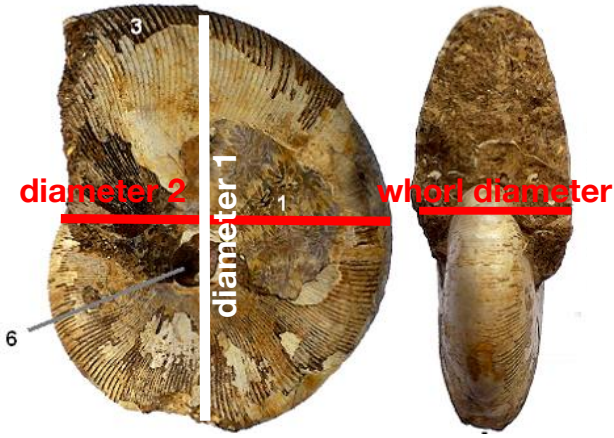
If valves are articulated

width
length_v
[do we also want length_d if articulated??]
height (if articulated)

If valves are disarticulated

width
length_v or length_d (if ventral or dorsal valve, respectively)
height_v or height_d (if ventral or dorsal valve, respectively)

Cephalopods

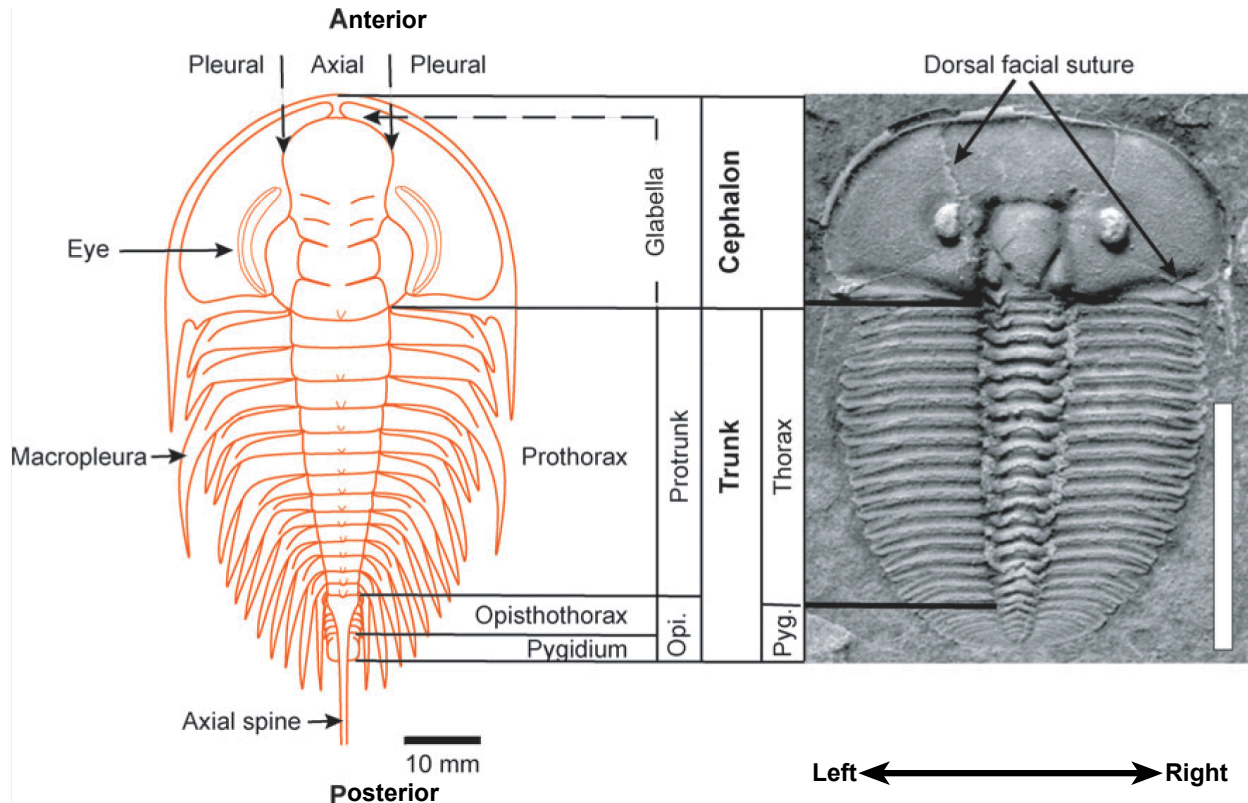


Cephalopod Measurements to Make
If Uncoiled or Heteromorphic
length_coiled
cone_diameter

If Coiled
diameter_1 (max diameter)
diameter_2
whorl_diameter

Trilobites

Measure as many things as you can. An articulated specimen like the one below should have measurements for the whole body, cephalon, and glabella.



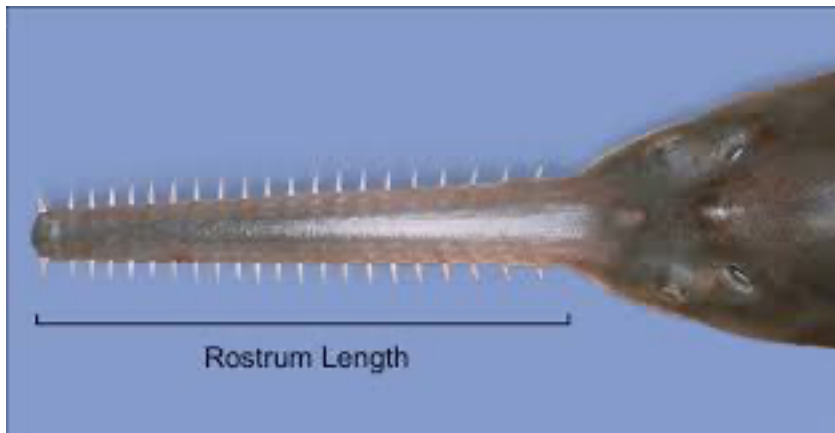
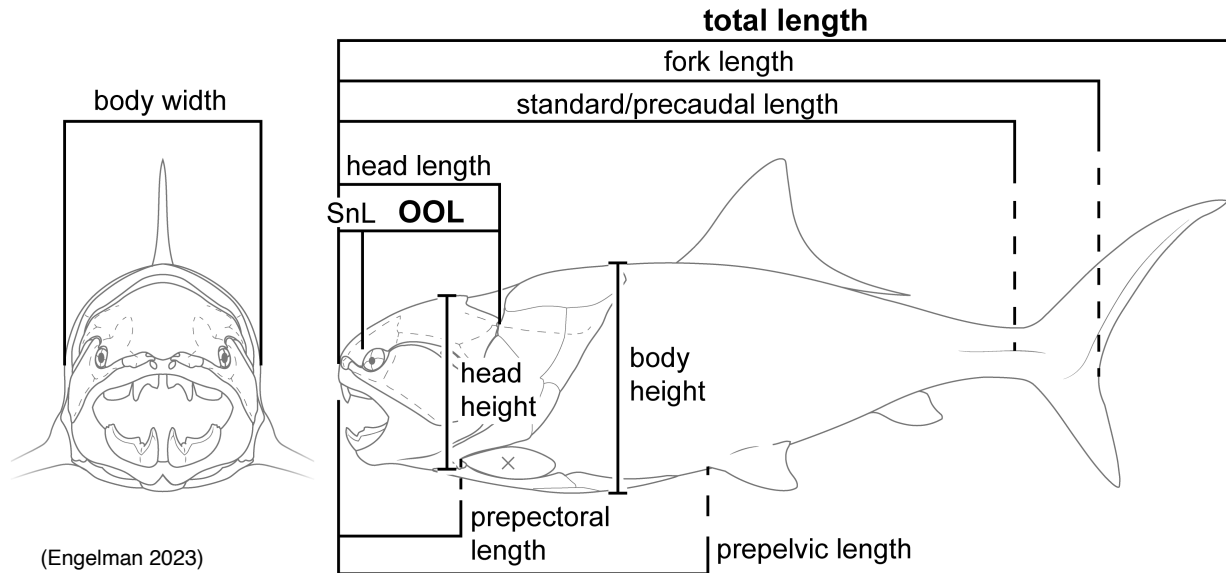
Watch out for disarticulated free cheeks when measuring cephalon width (right-left length). If cheeks missing, skip cephalon rl_length



Trilobite Measurements to Make (as many as possible)--dv_length will likely always be the height of the labella

- ap_length (trunk + cephalon)
- rl_length
- cephalon_ap_length
- cephalon_rl_length
- glabella_ap_length
- glabella_rl_length
- dv_length (if illustrated)
- pygidium_ap_length & pygidium_rl_length (if no cephalon)

Fish



Fish Measurements to Make

- Total Length
- Fork Length
- Standard Length
- Head Length
- Body Height
- Head Height
- Rostrum Length (if rostrum present)
- Body Width